BESS WorkBench — Summary

Design aid. Verify values against adopted codes and AHJ.
PDF body font is LiberationSans because Arial is not installed on this machine. The DOCX requests Arial.

Sample - 4 Room UPS BESS

SYNTHETIC PROJECT — not a real facility.
1 BLOCKED · 1 FAIL · 9 WARN · 2 INFO · 38 PASS
Ruleset ifc2024_nfpa855_2023 1.0.0

Fire areas

Fire area	Li-ion kWh	Status	Note
FA-1	600	WARN	Single fire area. A two-hour enclosure is the room minimum; three-hour separation is only required when a split is claimed.

Sprinklers and batteries

design area = min(room, 2500 sf) x 1; sprinkler gpm = 0.3 gpm/sf x design area; worksheet total = governing 240.00 gpm + hose 100.00 gpm
q = K * sqrt(p) = 5.60 * sqrt(7.00) = 14.82 gpm
FACP: (0.50 A x 24.00 h) + (2.00 A x 0.08 h) = 12.17 Ah; x 1.20 margin x 1.25 aging x 1.10 temp = 20.07 Ah
Gas panel: (0.25 A x 24.00 h) + (0.80 A x 2.00 h) = 7.60 Ah; x 1.20 margin x 1.25 aging x 1.10 temp = 12.54 Ah

Open items

ID	Assignee	Title
RFI-001	Controls vendor	Proof points and input shortfall
RFI-002	Mech	Setpoint conversion
RFI-003	Owner	Energy per fire area
RFI-004	Mech	Design LFL fraction
RFI-005	Owner	Sprinkler sign-off blocked without UL 9540A
RFI-006	Mech	Release mass-balance helper
RFI-007	Owner	UL 9540A report missing
RFI-008	Architect	Door hardware follows the barrier
RFI-009	Controls vendor	Sensor height still an open item
RFI-010	Controls vendor	Sensor height still an open item
RFI-011	Controls vendor	Sensor height still an open item
RFI-012	Controls vendor	Sensor height still an open item
RFI-013	Controls vendor	Sensor height still an open item
RFI-014	Controls vendor	Sensor height still an open item
RFI-015	Controls vendor	Sensor height still an open item
RFI-016	Controls vendor	Sensor height still an open item
RFI-017	Elec	High-gas signal to the FACP
RFI-018	Controls vendor	Relay landings not shown
RFI-019	Controls vendor	Monitoring-only or BACnet-only gas detection
RFI-020	Controls vendor	Single-gas percent LEL

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